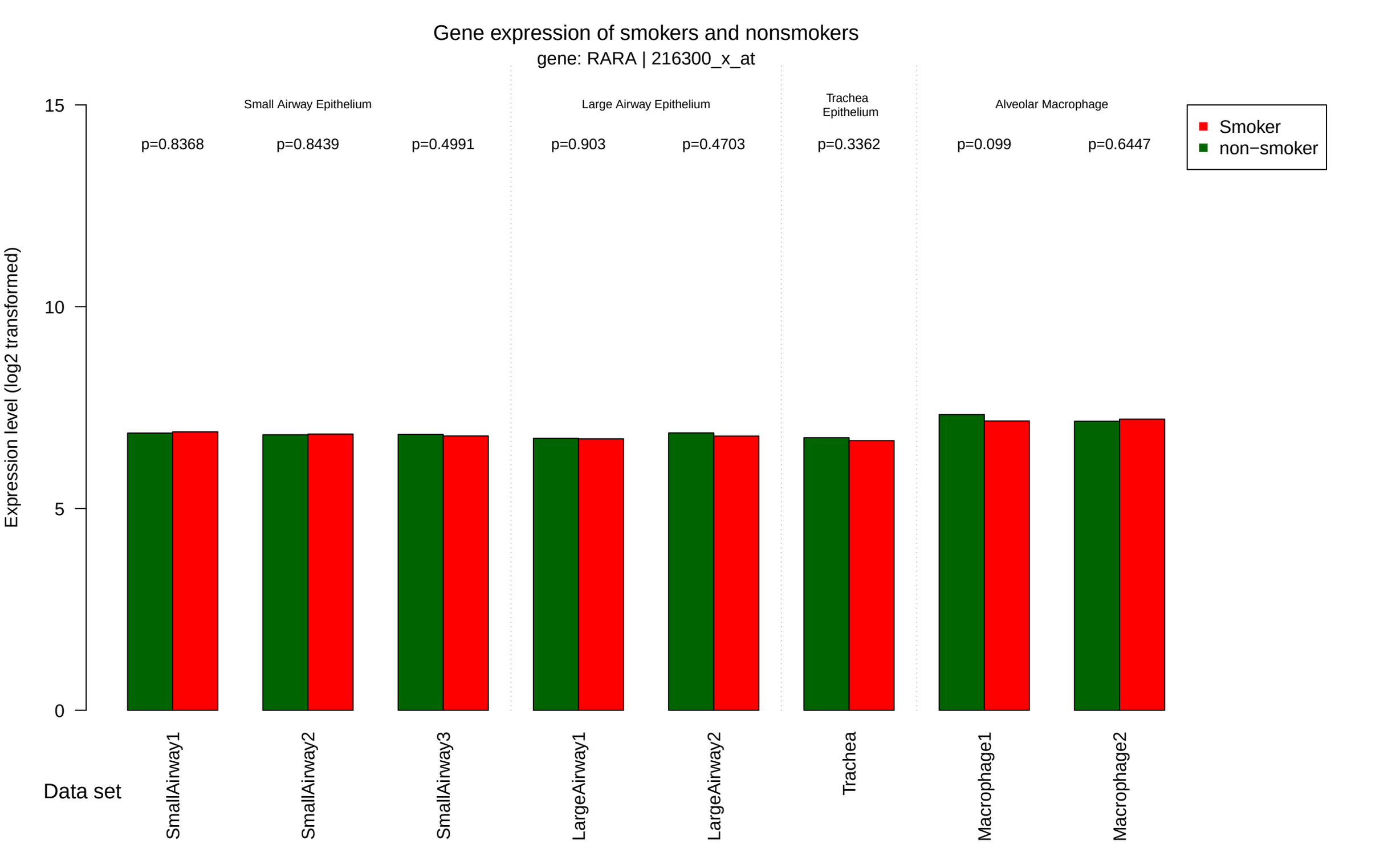
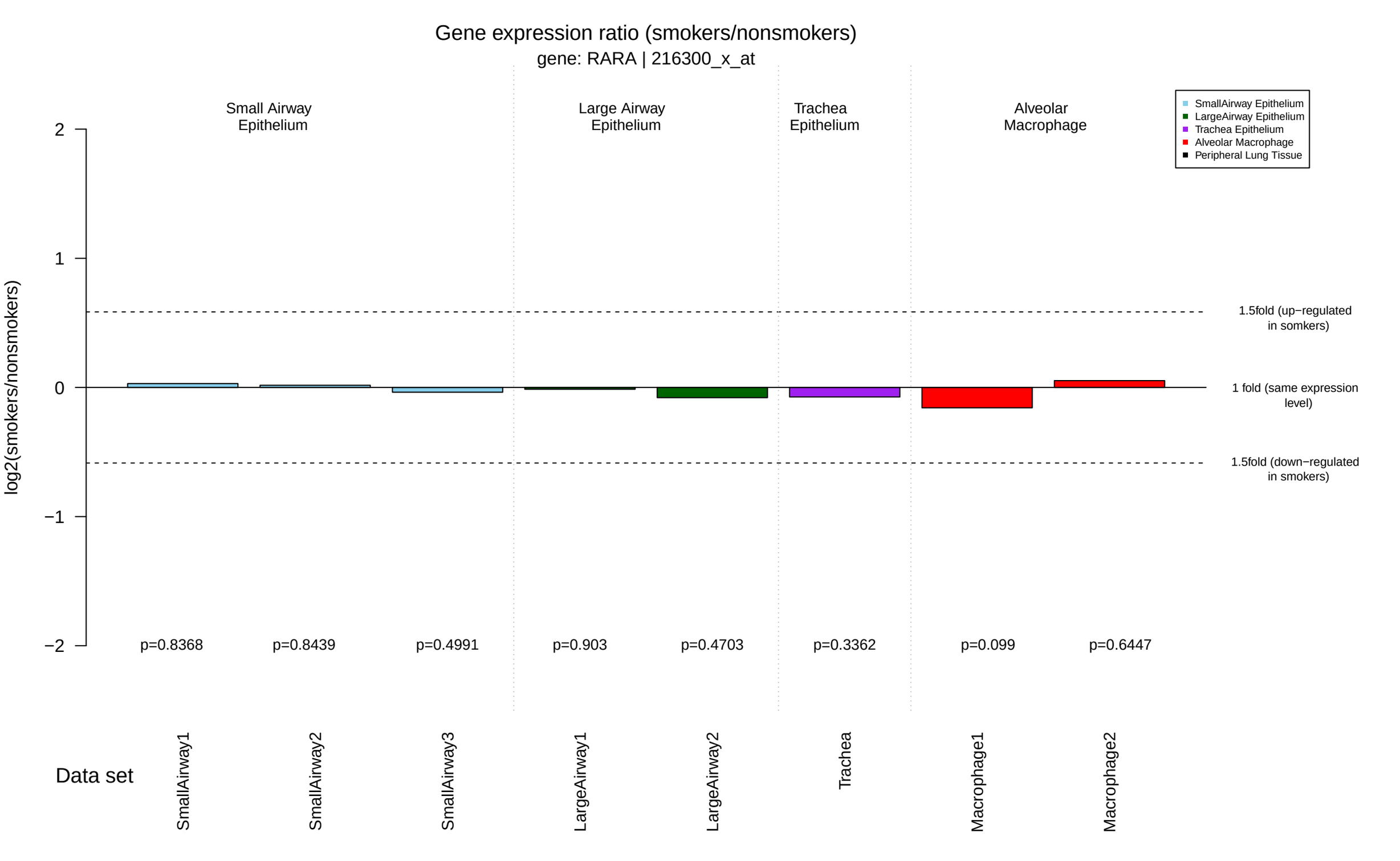
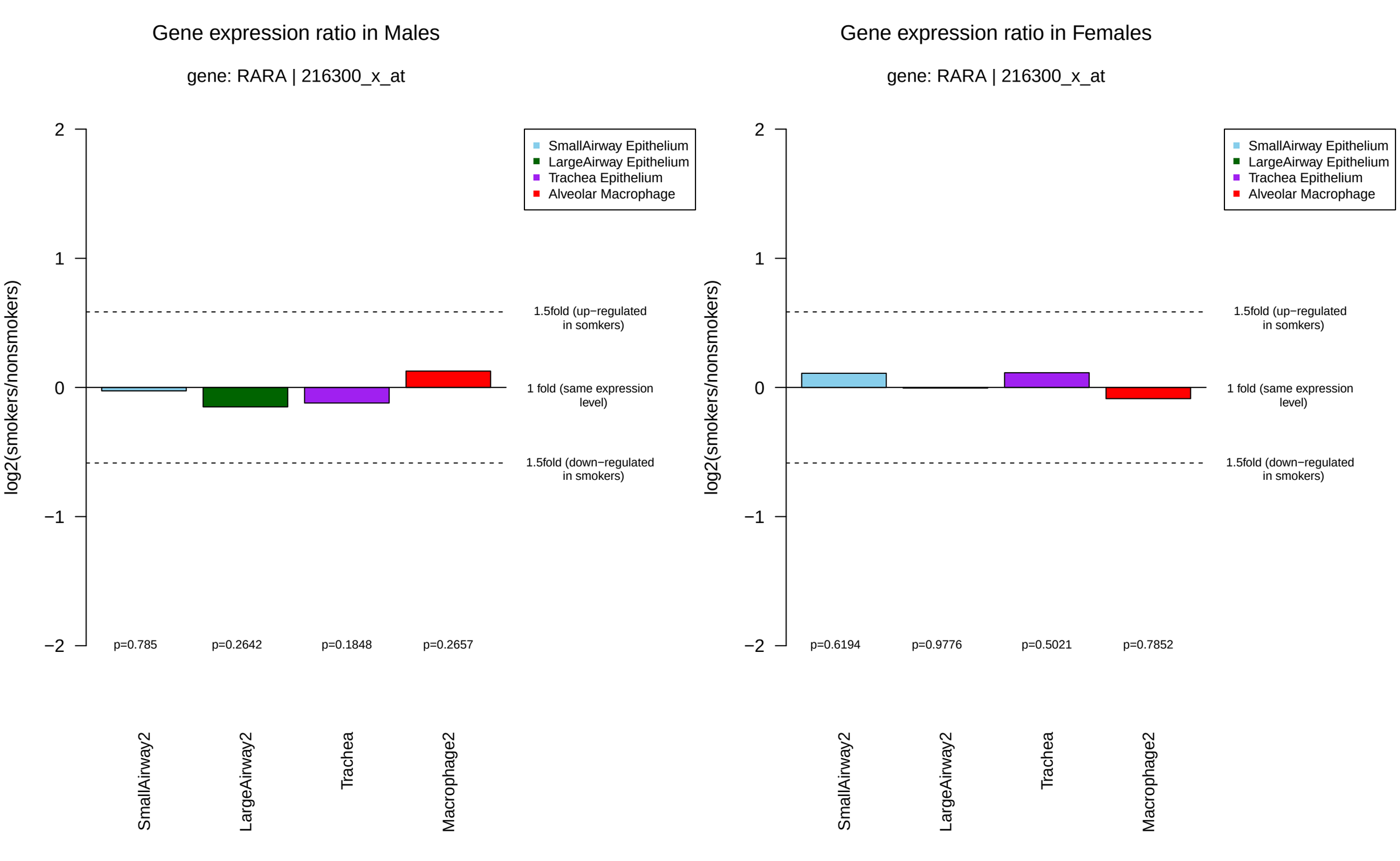
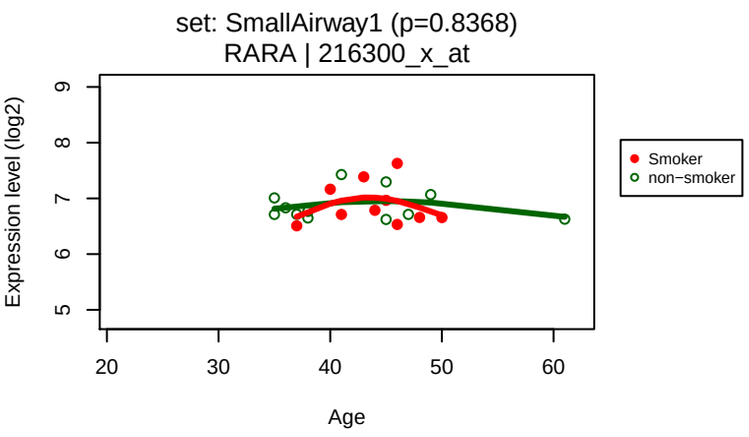
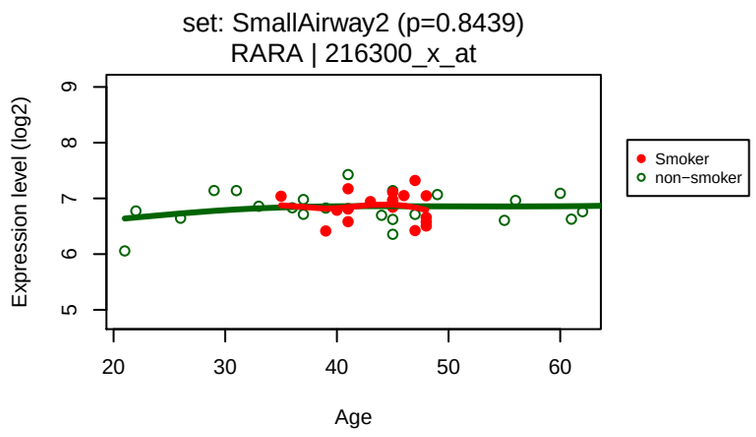
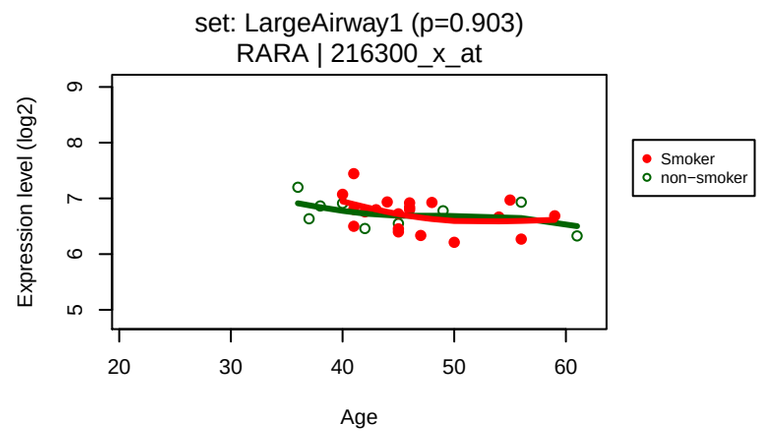

Gene expression of smokers and nonsmokers
gene: RARA | 216300_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8368
p=0.8439
p=0.4991
p=0.903
p=0.4703
p=0.3362
p=0.099
p=0.6447
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

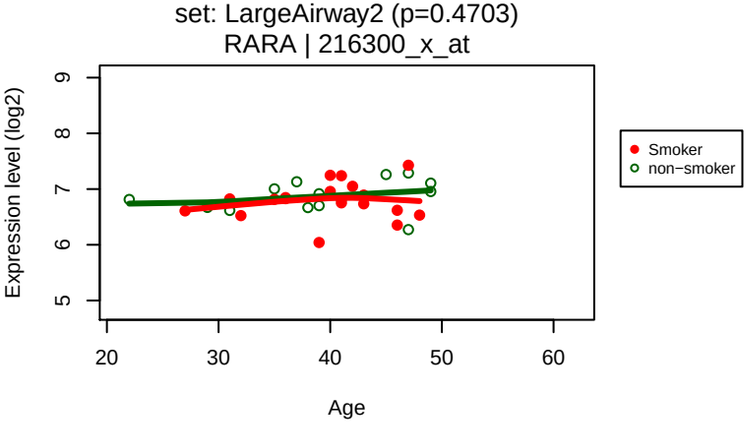

Gene expression ratio (smokers/nonsmokers)
gene: RARA | 216300_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8368
p=0.8439
p=0.4991
p=0.903
p=0.4703
p=0.3362
p=0.099
p=0.6447
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

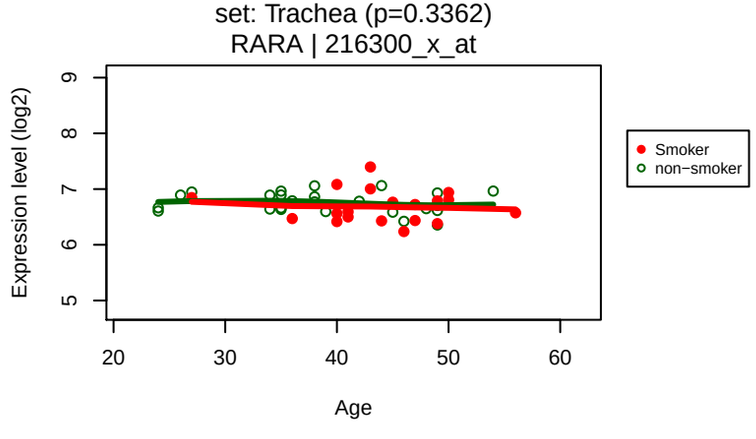

Gene expression ratio in Males
gene: RARA | 216300_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.785
p=0.2642
p=0.1848
p=0.2657
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARA | 216300_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6194
p=0.9776
p=0.5021
p=0.7852
SmallAirway2
LargeAirway2
Trachea
Macrophage2

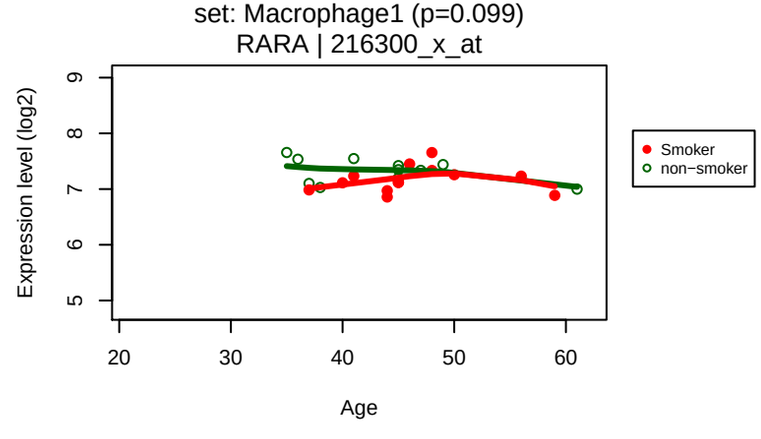

set: SmallAirway1 (p=0.8368)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker

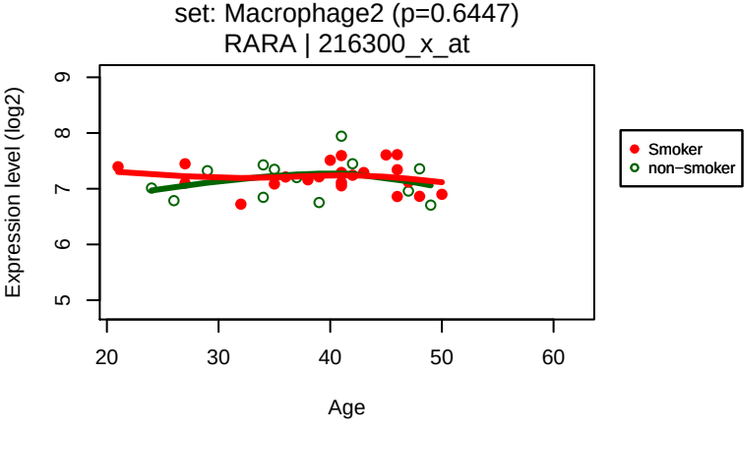

set: SmallAirway2 (p=0.8439)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker

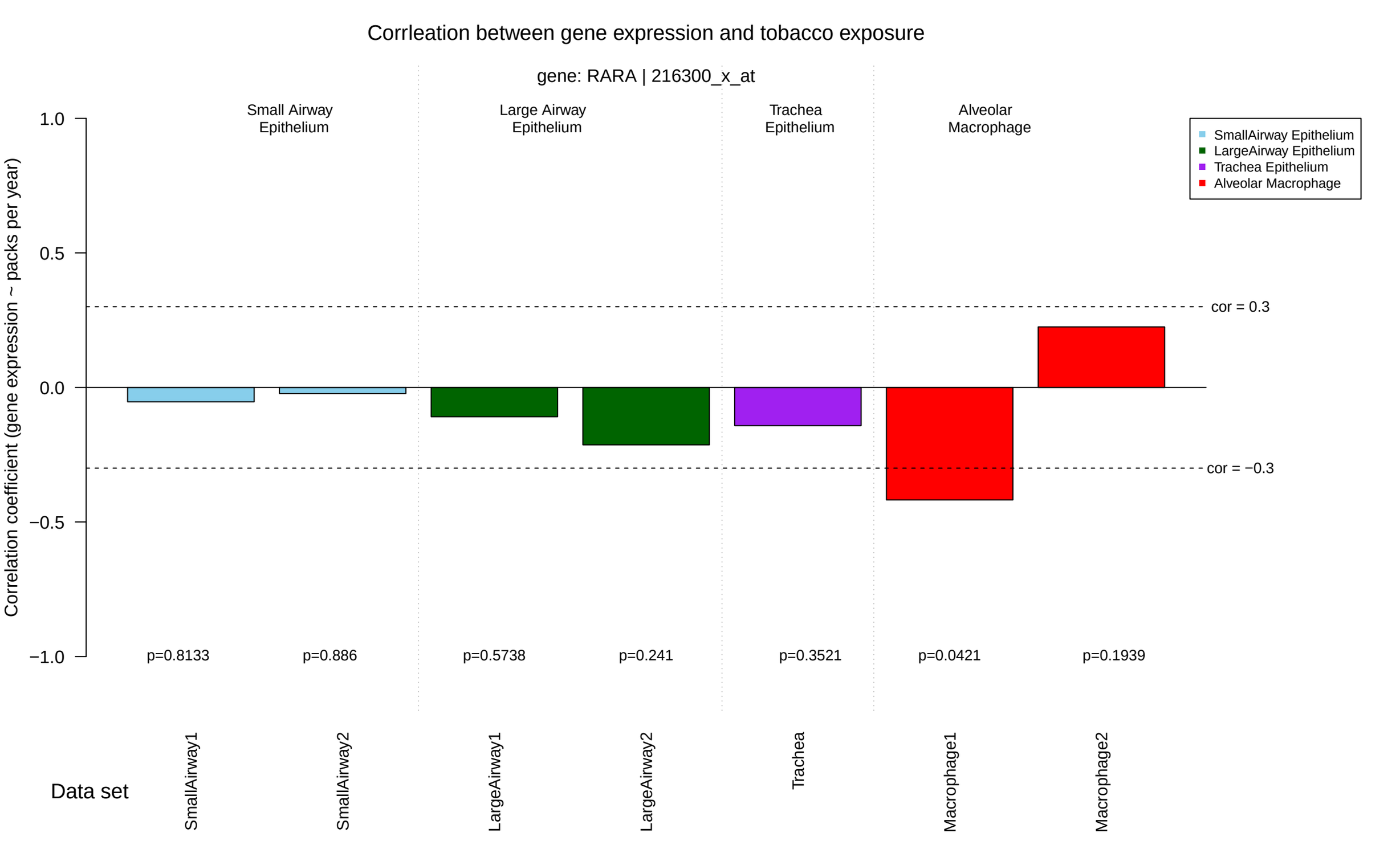

set: LargeAirway1 (p=0.903)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4703)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3362)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.099)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6447)
RARA | 216300_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARA | 216300_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8133
p=0.886
p=0.5738
p=0.241
p=0.3521
p=0.0421
p=0.1939
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2